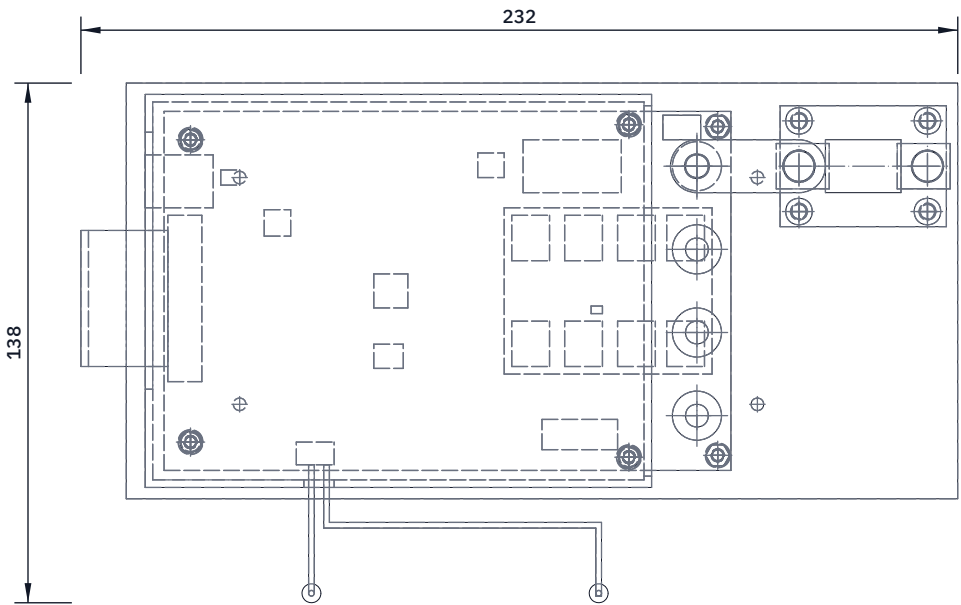


CONCEPT, NOT FOR FABRICATION

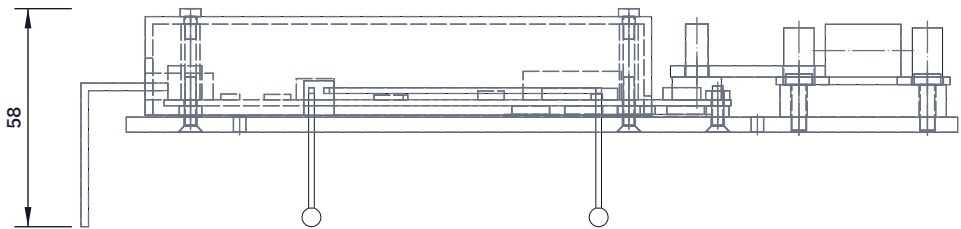
CellGuard · github.com/BoujeeEnjinia1701/cellguard

REV	DESCRIPTION	DATE	BY
P1	Preliminary GA (CGD-CAL-001, CGD-DDR-001)	2026-09-25	AC
P2	Made buildable: fixings, fuse link, cover notches (CGD-DDR-003)	2026-09-30	AC



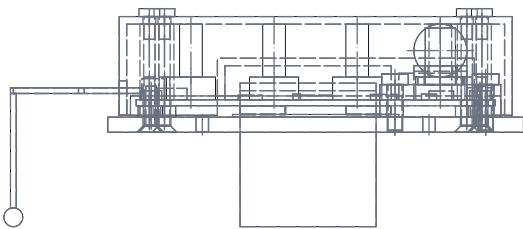
TOP VIEW

Scale 1:2; looking down (along -Z)



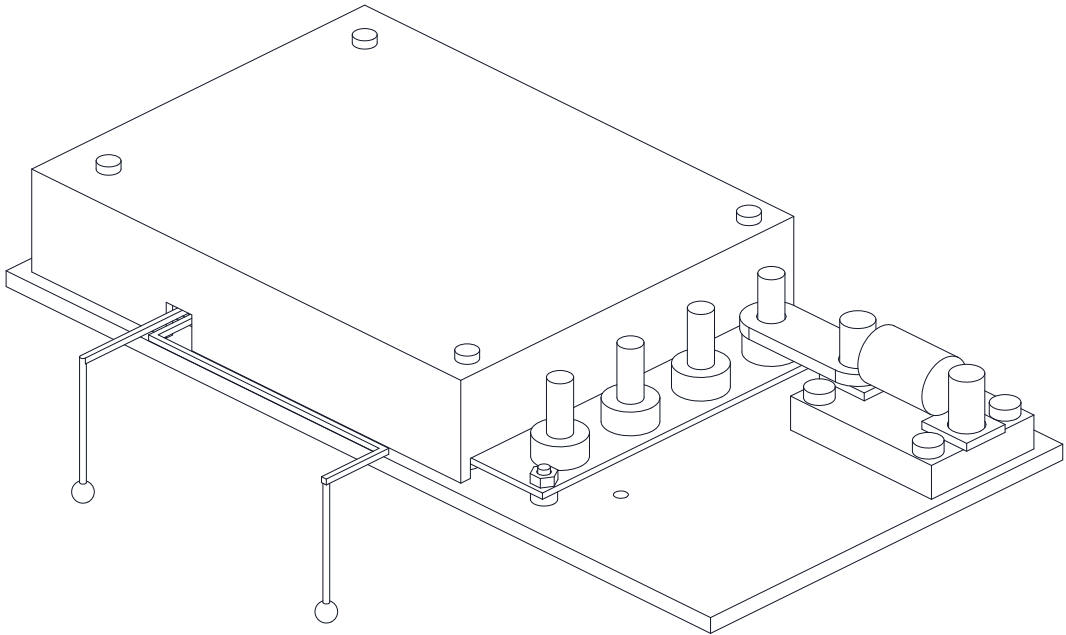
FRONT VIEW

Scale 1:2; looking at the front (along +Y)



RIGHT VIEW

Scale 1:2; looking at the right side (along -X)



ISOMETRIC VIEW

Not to scale; seen from the front right and above, about 30 deg elevation

KEY DIMENSIONS AND INTERFACES (MM)

- Board envelope 220 x 110 x 33 (harness tail and probes extra)
- Base plate 220 x 110 x 4, flat; board on 6 spacers 3 mm
- PCB fixings: M3 countersunk from below; 4 pillars 20 mm carry the cover
- PCB 150 x 95 x 1.6; top face at z 8.6
- Power studs M6 at x 66: B-, P-, P+, B+ at y -33, -11, 11, 33
- Pack fuse on plate in line with B+: 60 A, 80 V DC, 10 kA or more
- Copper link 14 x 3 joins the B+ stud to the fuse holder
- Plate mounting: 4 x M4 tapped, screws from below
- Balance: 17-way connector at -X; CAN and UART 6-way at -X
- MOSFETs 2 x 4 under PCB on gap pad, 1.0 mm pressed to 0.7
- Secondary protector and SCP fuse in the B+ path (part 14)
- CAN 2.0B 250 kbit/s, SwapCell v0.3 message set profile
- 40 A continuous, 80 A for 10 s; 4 to 16 LFP cells, under 60 V
- Mass about 0.66 kg (CGD-CAL-001 v0.3)
- PRELIMINARY, NOT FOR FABRICATION

CELLGUARD

github.com/BoujeeEnjinia1701/cellguard

General arrangement, BMS board

DRAWING NO. CGD-DWG-001			REV P2	SHEET 1 of 1	SIZE ANSI B
SCALE 1:2	UNITS mm	PROJECTION 	DATE 2026-09-30		DRAWN Amish Chadha
MATERIAL / NOTES Plate 6061 Al 4 mm; PCB FR-4 4-layer 2 oz; cover PC UL 94 V-0. See bom/bom.csv					
Licensed CERN-OHL-S-2.0 · Design Molecule Lab · designmolecule.com				Generated from cad/src; do not edit by hand	